

Chapter 50: Critical Care and Trauma — Revision Notes

Williams Obstetrics 26e, pp. 2291–2349 of the PDF. High-yield numbers are in **bold**. The nine tables are transcribed from the PDF images (they are missing from the extracted chapter text). 11 of the 12 figures are included from the PDF (Fig 50-8 repeats Fig 49-4 and is omitted).

Big picture: Critically ill gravidas are usually young and healthy, so the prognosis is generally good. Key themes: pulmonary edema in pregnancy is usually **noncardiogenic** (preeclampsia, sepsis, hemorrhage + fluids + tocolytics); ARDS needs **lung-protective ventilation** with a **PaO₂ >60 mm Hg** target to protect the fetus; sepsis needs **fluids + antibiotics within 1 h + source control**. In trauma, **stabilize the mother first**, then monitor the fetus (**≥4 h**) because **abruption can be occult**. In cardiac arrest: **left uterine displacement** and **perimortem cesarean by 4–5 min**.

1. Critical Care

- **1–3%** of US pregnant women need critical care each year; **risk of death during such admissions 2–11%**. Black women and Medicaid recipients are disproportionately affected.
- Greatest need: **hemorrhage, sepsis, hypertension**.
- Nonobstetrical antepartum admissions: diabetes, pneumonia/asthma, heart disease, chronic hypertension, pyelonephritis, thyrotoxicosis.
- **Cardiac disease = leading nonobstetrical indication for ICU admission**.
- Keep close to an **operating room** (hemorrhage; many are delivered preterm).

Organization of critical care

- ICUs from the 1960s; costly → **step-down intermediate care units** (between ICU and ward).

Table 50-1. Guidelines for Conditions That Could Qualify for Intermediate Care

System	Conditions
Cardiac	Evaluation for possible infarction, stable infarction, stable arrhythmias, mild to moderate congestive heart failure, hypertensive urgency without end-organ damage
Pulmonary	Stable patients for weaning and chronic ventilation; patients with potential for respiratory failure who are otherwise stable
Neurological	Stable central nervous system, neuromuscular or neurosurgical conditions that require close monitoring
Drug overdose	Hemodynamically stable
Gastrointestinal	Stable bleeding; liver failure with stable vital signs
Endocrinological	Diabetic ketoacidosis; thyrotoxicosis that requires frequent monitoring
Surgical	Postoperative from major procedures or complications that require close monitoring
Miscellaneous	Early sepsis; patients who require closely titrated IV fluids; pregnant women with severe preeclampsia or other medical problems

From the American College of Critical Care Medicine, 1998; Nasraway, 1998.

Obstetrical critical care — three unit types

1. **Medical/surgical ICU** (critical care specialists) — e.g., for **ventilatory support, invasive monitoring, pharmacological circulatory support**. About **0.5%** of obstetrical patients are transferred to these.

2. **Obstetrical intermediate care (high-dependency) unit** within labor and delivery — Parkland model; MFM, obstetricians and critical-care-trained nurses, with consultants as needed.
3. **Full obstetrical ICU** run by obstetrics and anesthesia near L&D (few centers).

Transport of the critically ill gravida

- Follow **EMTALA** for transfers.
- Minimum monitoring: **continuous pulse oximetry, ECG, regular vital signs**; secure IV access; confirmed and secured ETT if ventilated.
- **Left uterine displacement and supplemental O₂** routinely if antepartum. Fetal monitoring individualized.

Severe maternal morbidity (SMM)

- Population-based monitoring standard from **18 ICD-10 indicators** (CDC). Aim: find opportunities for prevention.
- **Limitations**: ignores comorbidities and referral case mix (e.g., accreta centers → more transfusions); indicators incomplete (only total, not supracervical, hysterectomy) → be cautious comparing hospitals.

Hemodynamic monitoring

- **Pulmonary artery catheter (PAC)** studies defined normal pregnancy hemodynamics and those of preeclampsia, ARDS and amniotic-fluid embolism — but **PAC is seldom necessary** now.
- **Echocardiography** is indispensable (anatomy, **RV function**).
- **NICOM** (noninvasive cardiac output monitoring) **overestimates** stroke volume and cardiac output vs Doppler echo; not widely adopted.

Table 50-2. Formulas for Deriving Various Cardiopulmonary Parameters

Parameter	Formula
Mean arterial pressure (MAP) (mm Hg)	$[\text{SBP} + 2(\text{DBP})] \div 3$
Cardiac output (CO) (L/min)	Heart rate $\times$ stroke volume
Stroke volume (SV) (mL/beat)	CO/HR
Stroke index (SI) (mL/beat/m ²)	Stroke volume/BSA
Cardiac index (CI) (L/min/m ²)	CO/BSA
Systemic vascular resistance (SVR) (dyn $\times$ s $\times$ cm ⁻⁵)	$[(\text{MAP} - \text{CVP})/\text{CO}] \times 80$
Pulmonary vascular resistance (PVR) (dyn $\times$ s $\times$ cm ⁻⁵)	$[(\text{MPAP} - \text{PCWP})/\text{CO}] \times 80$

BSA = body surface area (m²); CVP = central venous pressure (mm Hg); DBP = diastolic blood pressure; HR = heart rate (beats/min); MAP = mean systemic arterial pressure (mm Hg); MPAP = mean pulmonary artery pressure (mm Hg); PCWP = pulmonary capillary wedge pressure (mm Hg); SBP = systolic blood pressure.

- Values can be indexed to BSA; nonpregnant normal values are used, although they may not reflect the low-resistance uteroplacental circulation.

2. Acute Pulmonary Edema

- Incidence **~1 in 500 deliveries** at tertiary centers.

Type	Mechanism	In pregnancy
Noncardiogenic (permeability)	Capillary endothelial + alveolar epithelial damage	More common
Cardiogenic (hydrostatic)	High pulmonary capillary hydraulic pressure	Hypertension, hemorrhage/anemia, sepsis; diastolic dysfunction

- **>1/2** have sepsis with tocolysis, severe preeclampsia, or hemorrhage + vigorous fluids.

- One series (53 cases): **83% hypertensive disorders, 11% cardiac, 6% sepsis.**
- **β -mimetic tocolysis** once caused **up to 40%** of cases.

Noncardiogenic permeability edema

- Common denominator = **endotheliopathy** (preeclampsia, sepsis, hemorrhage — often combined).
- Often with **corticosteroids for fetal lungs + vigorous fluids + tocolysis**. Parenteral **β -agonists** clearly implicated; **MgSO₄** in some studies; combined therapy is causative.

Table 50-3. Some Causes and Associated Factors for Pulmonary Edema in Pregnancy

Type	Causes
Noncardiogenic permeability edema: endotheliopathy with capillary-alveolar leakage	Preeclampsia syndrome; acute hemorrhage; sepsis (pyelonephritis, metritis); tocolytic therapy (β -mimetics, MgSO ₄); aspiration pneumonitis; vigorous intravenous fluid therapy; pancreatitis
Cardiogenic pulmonary edema: myocardial failure with hydrostatic edema from excessive pulmonary capillary pressure	Hypertensive cardiomyopathy; obesity (adipositas cordis); left-sided valvular disease; vigorous intravenous fluid therapy; pulmonary hypertension

Cardiogenic hydrostatic edema

- Usually with hypertension, hemorrhage and anemia, or sepsis. **Diastolic dysfunction** from **chronic hypertension and/or obesity**; **acute systolic hypertension** tips it into edema.
- In preeclampsia with pulmonary edema, **half are undelivered**.

Diagnosis

- Clinical exam + **chest radiograph**; bedside **lung sonography** helps. Later echo often shows **normal ejection fraction**.
- **BNP: <100 pg/mL** — excellent NPV; **>500 pg/mL** — excellent PPV; **100–500** nondiagnostic (common). NT-proBNP and ANP are $\uparrow$ in **preeclampsia**.

Management

- **Emergency. Oxygen + furosemide 20–40 mg IV** as needed.
- O₂ delivery: nasal cannula and simple masks (imprecise, mix with room air); **nonrebreather** — **10–15 L/min**, FiO₂ approaching **95%**.
- Goal **SpO₂ >92%**.
- **Live fetus:** avoid cardioactive drugs that rapidly $\downarrow$ peripheral resistance (would $\downarrow$ uteroplacental flow).
- **Echocardiography** to define cardiogenic cause.
- **⚠ Acute pulmonary edema is NOT an indication for emergency cesarean.**

3. Acute Respiratory Distress Syndrome (ARDS)

- Acute lung injury $\rightarrow$ **severe permeability edema** and respiratory failure; a continuum from mild insufficiency to ventilator dependence.
- Incidence **60 ventilated cases per 100,000 births**; mortality **9%** (higher with sepsis). Antepartum ARDS $\rightarrow$ high perinatal mortality.

Definition

- **Radiographic pulmonary infiltrates + PaO₂:FiO₂ <200 + no heart failure.** Berlin definition: **mild / moderate / severe.**
- Clinical: dyspnea, tachypnea, desaturation.

Etiopathogenesis

- $>1/2$ of pregnant women with ARDS have sepsis, hemorrhage, shock and fluid overload in combination.
- Most frequent single causes: **pneumonia** (esp. **influenza**; also pneumococcus, metapneumovirus, Legionella) and **sepsis** (**pyelonephritis**, **puerperal pelvic infection**, **chorioamnionitis**). TRALI's contribution unclear.

Table 50-4. Some Causes of Acute Lung Injury and Respiratory Failure

Causes
Pneumonia: bacterial, viral, aspiration
Sepsis: chorioamnionitis, pyelonephritis, puerperal infection, septic abortion
Hemorrhage: shock, massive transfusion, transfusion-related acute lung injury (TRALI)
Preeclampsia syndrome
Tocolytic therapy
Embolism: amniotic fluid, trophoblastic disease, air, fat
Connective tissue disease
Substance abuse
Irritant inhalation and burns
Pancreatitis
Drug overdose
Fetal surgery
Trauma
Sickle-cell disease
Miliary tuberculosis
Cerebral hemorrhage

From Baron, 2018; Duarte, 2014; Lapinsky, 2015; Sibai, 2014; Snyder, 2013; Zeeman, 2006.

Three phases

1. **Exudative** — alveolar epithelial + microvascular endothelial injury → ↑ permeability, **surfactant loss**, ↓ lung volume, **shunting** → **hypoxemia**.
2. **Proliferative** — begins **~7 days**, lasts up to **21 days**; most recover here.
3. **Fibrotic** — healing. Long-term lung function is surprisingly good.

Clinical course

- Early: few signs (hyperventilation), adequate oxygenation → treat the underlying insult now.
- Progression: edema, ↓ compliance, ↑ shunting → dyspnea, tachypnea, hypoxemia; **bilateral** infiltrates.
- **Shunt >30%** → **refractory hypoxemia**; acidosis → arrhythmia, cardiac arrest.

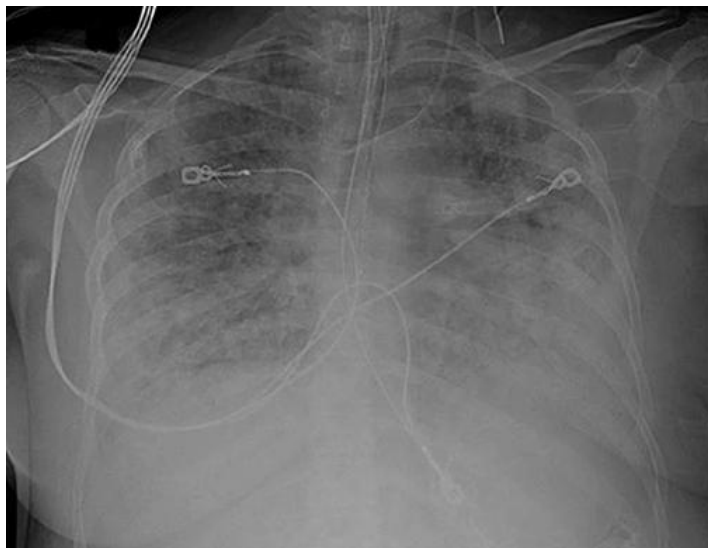


Figure 50-1. ARDS in the second trimester: marked bilateral parenchymal and pleural opacification.

Management

- Treat the underlying disorder; aggressive **antimicrobials** and **debridement**; support perfusion with crystalloid and blood.
- **Correct anemia** — the most efficient way to ↑ O_2 delivery: **1 g Hb carries 1.25 mL O_2** (at 90% saturation), whereas raising PaO_2 from 100 to 200 mm Hg adds only **0.1 mL O_2 per 100 mL blood**.
- **Goals: PaO_2 60 mm Hg or SaO_2 90%, with FiO_2 <50% and PEEP <15 mm Hg.**
- **Delivery to improve maternal oxygenation is controversial:** 10 of 29 ventilated women were delivered while intubated; ~half improved modestly; no predictors.

Mechanical ventilation

- **Noninvasive (face-mask) ventilation** may help early.
- **Intubate early** if respiratory failure is likely or imminent. Because $PaCO_2$ is normally low in pregnancy, **consider intubation when $PaCO_2$ reaches 35–40 mm Hg**.
- **Tidal volume ≤ 6 mL/kg** initially.
- **Targets: PaO_2 >60 mm Hg or $SaO_2 \geq 90\%$; $PaCO_2$ 35–45 mm Hg.** Lower PaO_2 may impair placental perfusion.
- Modes: **assist-control, intermittent mandatory, pressure-support**; CPAP for those needing little support.
- Pregnancy series: 51 ARDS cases — half had severe preeclampsia; 2 maternal deaths. 29 ventilated women — mortality **10%**; delivery while ventilated gave little respiratory benefit.
- **PEEP 5–15 mm Hg** is usually safe and recruits collapsed alveoli; higher levels → ↓ venous return, ↓ cardiac output, ↓ uteroplacental perfusion, overdistention, **barotrauma**.
- **ECMO (veno-venous):** influenza lung failure (12 women; 3 of 4 deaths from **anticoagulation-related hemorrhage**); maternal and perinatal mortality **~30%**; maternal survival lower than nonpregnant.

Fetal oxygenation

- Rightward shift of the O_2 –Hb curve (↓ affinity, better tissue unloading): **hypercapnia, metabolic acidosis, fever, ↑ 2,3-DPG. 2,3-DPG rises ~30%** in pregnancy.
- **Fetal Hb curve lies to the left: $P50 = 27$ mm Hg (maternal) vs 19 mm Hg (fetal).** The fetus is always on the steep dissociation part of the curve, so O_2 transfer to fetal tissues is favored even with low maternal PaO_2 (high altitude: maternal PaO_2 60 mm Hg, fetal PaO_2 as at sea level).

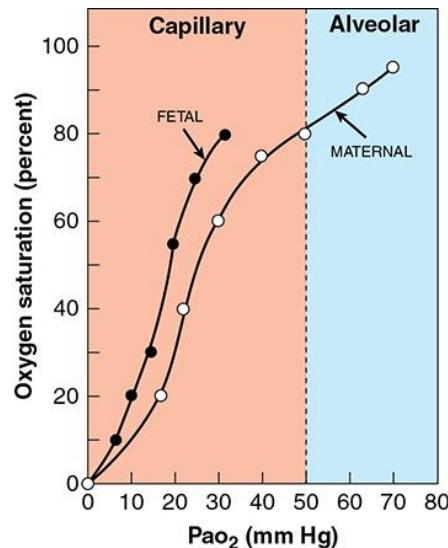


Figure 50-2. Oxyhemoglobin dissociation: the fetal curve lies left of the maternal curve, so fetal Hb is more saturated at any PaO_2 (P_{50} 19 vs 27 mm Hg).

Intravenous fluids

- **Conservative** fluid strategy → fewer ventilator days (same mortality).
- **Colloid oncotic pressure (COP): 1 g/dL albumin $\approx$ 6 mm Hg.** COP falls **28 (nonpregnant) → 23 (term) → 17 mm Hg (puerperium)**; lower in preeclampsia.
- **COP–wedge gradient:** normally **>8 mm Hg**; **≤ 4 mm Hg** → **↑ pulmonary edema risk**.
- **Albumin offers no benefit over crystalloid.**

Long-term outcomes

- Near-normal lung function, but impaired **cognition** at 3–12 months, delayed return to activity (**1–2 years**), exercise limitation and psychological sequelae at 5 years (nonpregnant data).

4. Sepsis and Septic Shock

Definitions (Sepsis-3)

- **Sepsis = infection + acute organ dysfunction.**
- **Septic shock = hypotension unresponsive to fluids — needs vasopressors despite fluid resuscitation** (nonpregnant mortality **$\sim 20\%$**).
- **"SIRS" and "severe sepsis" are no longer recommended terms.**
- Incidence **2.4 per 10,000** deliveries (55 million births); SMFM cites **4–10 per 10,000**. Infection/sepsis caused **12%** of US pregnancy-related deaths (2013–2017).
- Case fatality is **lower in pregnant** than age-matched nonpregnant women, but maternal sepsis mortality is underestimated.

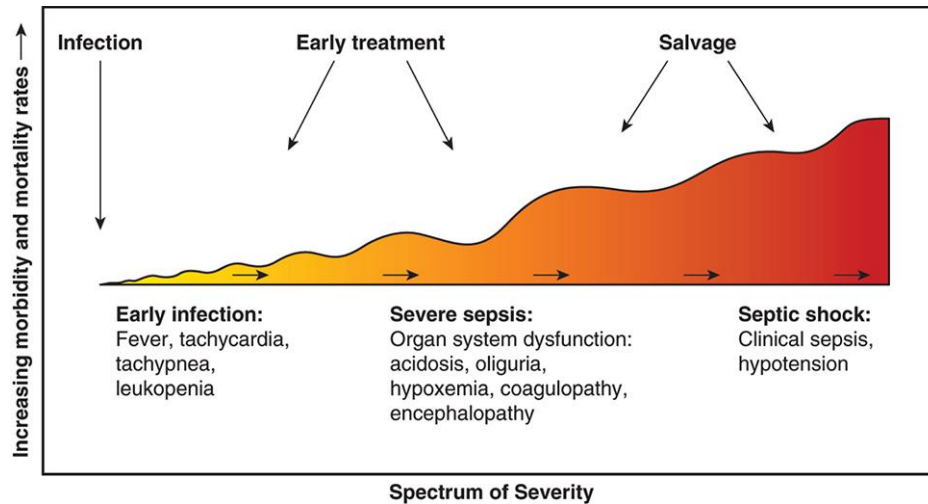


Figure 50-3. Sepsis continuum: early infection (fever, tachycardia, tachypnea, leukopenia) → organ dysfunction (acidosis, oliguria, hypoxemia, coagulopathy, encephalopathy) → septic shock; early treatment beats salvage.

Common obstetrical sources: pyelonephritis, chorioamnionitis and puerperal sepsis, pneumonia, septic abortion, necrotizing fasciitis.

Etiopathogenesis

- **LPS (endotoxin)** of gram-negative bacteria — **lipid A** binds mononuclear cells → cytokine release.
- **Pyelonephritis:** *E. coli*, *Klebsiella*. Pelvic infections are polymicrobial; severe sepsis most often from **Enterobacteriaceae (*E. coli*)**; also aerobic/anaerobic streptococci, *Bacteroides*, *Clostridium*.
- **Superantigens** (group A β -hemolytic streptococci, *S. aureus* incl. **CA-MRSA**) activate T cells → **toxic shock syndrome**.
- **Exotoxins:** *Clostridium perfringens* / *sordellii*, **TSST-1** (*S. aureus*), streptococcal toxic shock-like exotoxin → rapid **tissue necrosis and gangrene** (esp. postpartum uterus), cardiovascular collapse; maternal mortality **58%** (Yamada review).
- **"Cytokine storm"** (TNF- α , interleukins, proteases, oxidants, bradykinin) → **selective vasodilation**, maldistributed flow, leukocyte/platelet **capillary plugging, capillary leak**. **IL-6** likely mediates **myocardial suppression**.
- Early shock = $\downarrow$ **SVR not fully compensated by $\uparrow$ CO** → lactic acidosis, $\downarrow$ O_2 extraction, organ injury (lung, kidney); **septic cardiomyopathy** common.

Table 50-5. Clinical Manifestations of Sepsis

System	Manifestations
Central nervous system	Confusion, delirium, somnolence, coma, combativeness, fever, hypothermia
Cardiovascular	Tachycardia, hypotension
Pulmonary	Tachypnea, arteriovenous shunting with dysoxia and hypoxemia, exudative infiltrates from endothelial-alveolar damage, pulmonary hypertension
Gastrointestinal	Gastroenteritis — nausea, vomiting and diarrhea; ileus; hepatocellular necrosis — jaundice, transaminitis
Renal	Prerenal oliguria, azotemia, acute kidney injury, proteinuria
Hematological	Leukocytosis or leukopenia, thrombocytopenia, activation of coagulation with disseminated intravascular coagulation
Endocrinological	Hyperglycemia, adrenal insufficiency
Cutaneous	Acrocyanosis, erythroderma, bullae, digital gangrene

Hemodynamic phases

- **Warm phase** (after crystalloid): **high cardiac output, low SVR**, pulmonary hypertension, myocardial depression.
- Good response to fluids + antibiotics ± debridement = favorable. If shock persists despite fluids and **β-adrenergic inotropes** fail → severe leak and/or myocardial depression.
- **Cold phase: oliguria + peripheral vasoconstriction** — rarely survived.
- Continued renal, pulmonary, cerebral dysfunction after BP correction = poor sign. **Each failed organ system adds 15–20% to the risk of death.**

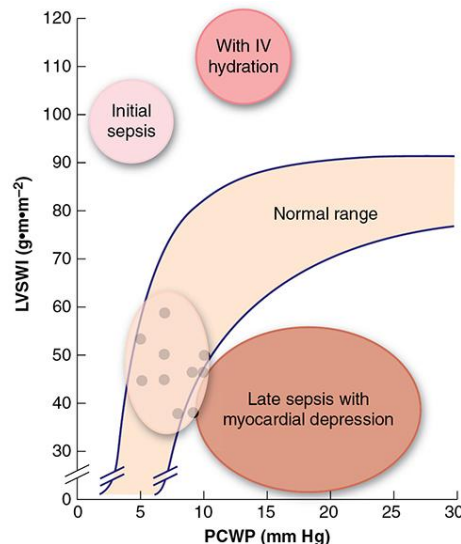


Figure 50-4. Ventricular function in sepsis: early sepsis has high LVSWI at low PCWP (↑ further with IV fluids); late sepsis shows myocardial depression (low LVSWI, high PCWP).

Management

- **Early goal-directed therapy (EGDT)** — no survival benefit in 3 trials → **no longer recommended**, though its tenets remain.
- Obstetrical early warning scores (**MEWS**) predict sepsis poorly (PPV **1–2%**); **Sepsis in Obstetrics Score** under study.
- **Core steps:** broad-spectrum antibiotics, **cultures, serum lactate, IV fluid bolus, vasopressors as needed** — performed **simultaneously** (Evaluate, Assess, Manage).

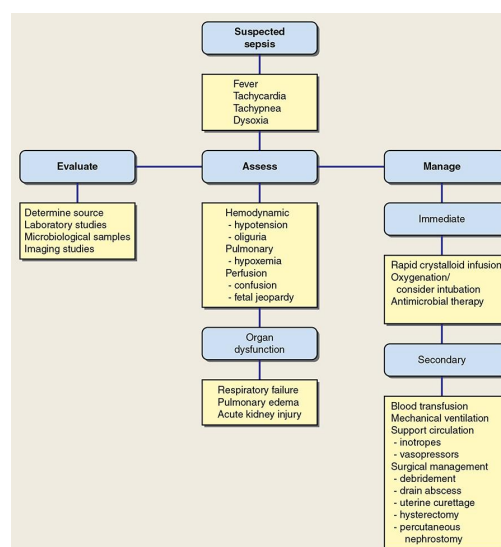


Figure 50-5. Sepsis algorithm: evaluate the source, assess hemodynamic/pulmonary/perfusion status and organ dysfunction, and manage immediately (crystalloid, oxygenation, antimicrobials) then secondarily (blood, ventilation, inotropes/vasopressors, surgery).

- **Most important step: rapid crystalloid 2 L** (sometimes **4–6 L**) to restore renal perfusion.

- **Broad-spectrum antibiotics within 1 hour**, maximal doses, after cultures.
 - **Pelvic infection: ampicillin + gentamicin + clindamycin**; add **vancomycin** for likely **MRSA** (incisional, soft tissue).
 - Pyelonephritis: cover Enterobacteriaceae.
 - Septic abortion or deep fascial infection: **Gram stain** for *Clostridium* or group A strep.
- **Blood** if anemic (hemoconcentration masks it). Nonpregnant ICU: Hb ≥ 9 vs ≥ 7 g/dL no different — but higher Hb improves fetal oxygenation.
- **Colloids**: hydroxyethyl starch $\rightarrow$ $\uparrow$ mortality in one trial, equivalent in another; **albumin not superior** to crystalloid.
- Target **urine output ≥ 30 , preferably 50 mL/h**; if not $\rightarrow$ vasoactive drugs.
- Alveolar flooding may occur with normal wedge pressure $\rightarrow$ ventilation.

Surgical source control

- **Debride necrotic tissue, drain pus** — crucial.
- **Septic abortion $\rightarrow$ prompt uterine evacuation**; hysterectomy only for **gangrene**.
- **Pyelonephritis with ongoing sepsis** $\rightarrow$ look for **obstruction** (calculi, perinephric/intrarenal phlegmon or abscess): renal sonography, "one-shot" IVP, CT/MRI $\rightarrow$ **ureteral stent, percutaneous nephrostomy**, or flank exploration.

Puerperal infections — four exceptions to "antibiotics alone cure"

1. **Uterine myonecrosis/gangrene** from **group A β -hemolytic streptococcus** or **clostridia** $\rightarrow$ high mortality; **prompt hysterectomy may be lifesaving**. If toxic shock without uterine necrosis (excluded by **CT**), hysterectomy may be unnecessary.
2. **Necrotizing fasciitis** of episiotomy or abdominal incision $\rightarrow$ **surgical emergency** (Fournier gangrene reported).
3. **Uterine incisional necrosis / dehiscence with peritonitis** (later onset, less aggressive) $\rightarrow$ CT; hysterectomy usually needed.
4. Ruptured **parametrial, intraabdominal or ovarian abscess** (less common).

Table 50-6. Clinical Findings in 126 Women with Group A β -Hemolytic Streptococcal Infection

Findings
High fever
Respiratory symptoms
Gastrointestinal symptoms
Uterine hypertonus
Uterine tenderness
Capillary leak — dyspnea, cough; pulmonary edema; bloating, distention
Shock

Data from Kaiser, 2018; Yamada, 2010.

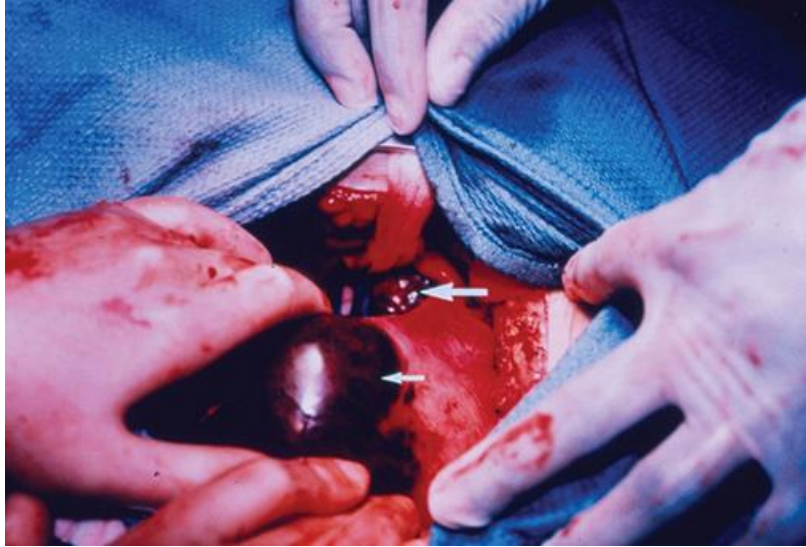


Figure 50-6. Fatal group A streptococcal puerperal sepsis after an uncomplicated term vaginal birth: black, ballooned gangrenous areas of the uterus (arrows) at hysterectomy.

Adjunctive therapy

- **Vasopressors only after aggressive fluids fail. First-line: norepinephrine and vasopressin;** norepinephrine is best studied in pregnancy (SMFM).
- **Low-dose corticosteroids** may be considered in septic shock (**hydrocortisone + fludrocortisone** ↓ 90-day mortality; Annane 2018).
- Endotoxin ↑ tissue factor (procoagulant) and ↓ activated protein C; anticoagulant agents **do not improve outcomes**.
- **Induced hypothermia does not help.**

5. Trauma

- **10–20%** of gravidas sustain physical trauma. **Injury is the most common identified nonobstetrical cause of maternal death;** pregnant trauma victims are **2x** as likely to die.
- Parkland: **motor vehicle accidents + falls = 85%** of injuries.
- Pregnancy-associated **suicide 2 / 100,000** and **homicide 3 / 100,000** live births (often linked to IPV).

Intimate partner violence (IPV)

- Physical, sexual or psychological harm by a current/former partner; affects **1 in 5 women** in their lifetime; continues in pregnancy.
- Linked to poverty, low education, tobacco, alcohol, drugs. **Prior IPV** is a major risk factor for **intimate partner homicide**. More common in women seeking termination.
- Outcomes: uterine rupture, preterm delivery, maternal and perinatal death; later abortion, preterm/LBW. **~3x perinatal death** (metaanalysis).
- **Screen universally: first prenatal visit, each trimester, and postpartum.** Women who refuse to answer may be at highest risk; abused women present late.

Sexual assault

- Lifetime prevalence **20%**; **2%** of victims (Dallas) were pregnant. Physical trauma is common; **evidence collection protocol is unchanged**.
- **Prophylaxis against gonorrhea, chlamydia, trichomoniasis** (CDC) ± HIV; emergency contraception if not pregnant.
- **PTSD, major depression, suicidal ideation: 30–35% lifetime risk each** → counseling.

Table 50-7. Guidelines for Prophylaxis Against Sexually Transmitted Disease in Pregnant Victims of Sexual Assault

Prophylaxis Against	Regimen ^a	Alternative
<i>Neisseria gonorrhoeae</i>	Ceftriaxone 250 mg IM single dose PLUS azithromycin 1 g single dose	Cefixime 400 mg single dose PLUS azithromycin 1 g single dose
<i>Chlamydia trachomatis</i>	Azithromycin 1 g single dose^b OR amoxicillin 500 mg three times daily for 7 days	Erythromycin-base 500 mg four times daily for 7 days
<i>Trichomonas vaginalis</i>	Metronidazole 2 g single dose OR tinidazole 2 g single dose	Metronidazole 500 mg twice daily for 7 days
Hepatitis B (HBV)	If not previously vaccinated, give first dose HBV vaccine IM, repeat at 1–2 and 4–6 months	—
Human immunodeficiency virus (HIV)	If exposure ≤ 72 h ago, consider antiretroviral prophylaxis if risk for HIV exposure is high	Tenofovir disoproxil fumarate/emtricitabine (Truvada) once daily plus raltegravir 400 mg (Isentress) twice daily for 28 days

^aOral administration unless otherwise stated. ^bFor nonpregnant women, doxycycline 100 mg orally twice daily for 7 days can be given instead. IM = intramuscularly. From Centers for Disease Control and Prevention, 2015, 2016.

Automobile accidents

- **$\geq 3\%$** of pregnant women per year; **90% of nonviolent blunt trauma**; the most common cause of serious/fatal blunt trauma and the **leading cause of traumatic fetal death**. Alcohol often involved.
- **Up to $\frac{1}{2}$ occur without seat belts**. **Three-point restraint: lap belt below the uterus across the upper thighs**, shoulder belt above the uterus $\rightarrow$ prevents steering-wheel contact and $\downarrow$ abdominal impact pressure.
- **Airbags**: perinatal outcomes not different (2207 vs 1141; **96%** belted) $\rightarrow$ injuries relate to crash severity.



Figure 50-7. Correct three-point restraint: shoulder belt between the breasts and above the uterus; lap belt snug across the upper thighs, well below the uterus.

Other blunt trauma

- Falls and assaults ($\approx \frac{1}{3}$ of hospitalized trauma intentional). **Bowel injury less frequent** (uterus protects), but diaphragm, spleen, liver, kidney injured; **retroperitoneal hemorrhage** more common. **Amnionic-fluid embolism** even after mild trauma.
- **Orthopedic injuries: 6%** (Parkland) $\rightarrow$ $\uparrow$ abruption, preterm delivery, perinatal death.
- **Pelvic fracture**: maternal mortality **9%**, fetal **35%**.

Fetal injury and death

- Rises with maternal injury severity; more likely with **direct fetoplacental injury, maternal shock, pelvic fracture, maternal head injury, hypoxia**.
- **Motor vehicle accidents** → 82% of traumatic fetal deaths; **placental injury in 1/2, uterine rupture 4%**.
- Fetal skull/brain injury: **engaged head + maternal pelvic fracture**; contrecoup head injury in unengaged or nonvertex fetuses. **Fetal skull fractures best seen by CT** (see Fig 49-4).

Placental injuries

- **Traumatic abruption**: elastic myometrium deforms around the **inelastic placenta** (deceleration against wheel or belt, or **coup–contrecoup**).
- **Abruption in 1–6% of minor and up to 50% of major injuries**; more likely at **>30 mph**.
- ⚠ **May be occult**: Parkland 13 cases — **11 uterine tenderness, only 5 vaginal bleeding**. More often **concealed**, with higher intrauterine pressure → **coagulopathy more common** than with spontaneous abruption.
- Warning signs: uterine activity, **fetal tachycardia, sinusoidal pattern, late decelerations**, acidosis, fetal death. CT is accurate but routine CT screening for abruption is not advocated.
- **Placental tear** (linear or stellate) → life-threatening **fetal hemorrhage** into amnion or maternal circulation.
- **Fetomaternal hemorrhage**: small FMH in up to 1/3 of trauma cases, **<15 mL in 90%**. Evidence of FMH → **20× risk of contractions/preterm labor**. Severe fetal bleeding → long-term neurological harm.

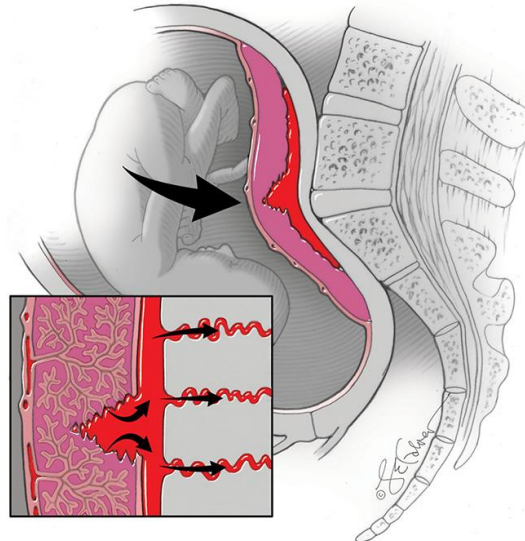


Figure 50-9. Coup–contrecoup: the inelastic placenta "fractures" against the sacral promontory; retroplacental blood can be forced into maternal venules (fetomaternal hemorrhage — Kleihauer–Betke).

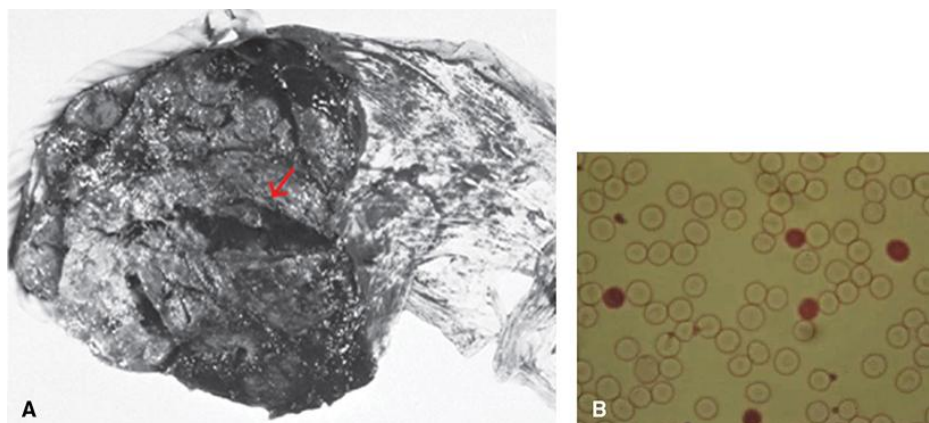


Figure 50-10. (A) Placental laceration (arrow) under a partial abruption caused fatal fetomaternal hemorrhage; (B) Kleihauer–Betke smear — dark fetal cells made up 4.5% of maternal RBCs.

Uterine rupture

- **<1%** of severe blunt trauma; more likely with a **scarred uterus** and direct high-force impact. A **25-mph** collision can generate **500 mm Hg** intrauterine pressure in a restrained woman.

Penetrating trauma

- **9%** of abdominal trauma in pregnancy: **77% gunshot, 23% stab**.
- **Maternal visceral injury only 15–40%** (vs 80–90% nonpregnant) — the uterus shields; **fetus injured in ⅓**, maternal viscera **20%**.
- vs blunt injury: maternal mortality **7% vs 2%**, fetal mortality **73% vs 10%**.

Management of trauma

- Injury severity scores **do not predict** pregnancy outcome; **minor injuries** can cause preterm labor and abruption. Minor trauma ≥ 24 wk: **14%** had contractions needing monitoring **>4 h**.
- **Priorities: evaluate and stabilize the mother first** (fetal assessment must not distract). Ventilation, stop hemorrhage, treat hypovolemia with crystalloid and blood; **displace the uterus off the great vessels** after midpregnancy.
- Then search for fractures, internal injuries and uterine/placental/fetal trauma. **Radiography is not proscribed** when indicated. Screening sonography $\rightarrow$ CT if positive.
- **FAST scan** (focused assessment with sonography for trauma): **5-minute, 4–6 views** — perihepatic, perisplenic, pelvic, pericardial. Upper-abdominal fluid implies **>500 mL hemoperitoneum** (not validated in pregnancy). Open peritoneal lavage sometimes.



Figure 50-11. FAST scan: free fluid (blood, marked by the asterisk) in Morison pouch between liver and kidney; this patient had 2500 mL of hemoperitoneum.

- **Penetrating injury:** peritoneal signs are blunted $\rightarrow$ **aggressive exploratory laparotomy. Mandatory for abdominal gunshot wounds**; selected stab wounds may be observed; diagnostic laparoscopy used.
- **Cesarean:** laparotomy itself is **not** an indication; depends on GA, fetal condition, uterine injury, and whether the uterus hinders repair.

Electronic fetal monitoring after trauma

- Fetal status is a maternal "vital sign"; may reveal abruption even if the mother is stable.
- **Contractions less often than every 10 min over 4 h $\rightarrow$ no abruption; more often $\rightarrow$ ~20% abruption** (Pearlman 1990).
- **Normal tracing $\rightarrow$ no adverse outcomes.** Avoid **tocolytics** (mask abruption).

- **Start monitoring once the mother is stable; observe ≥ 4 hours** if normal. **Continue** with contractions, nonreassuring FHR, bleeding, uterine tenderness/irritability, serious maternal injury or ruptured membranes. Rarely abruption occurs **days later**.

Fetal-maternal hemorrhage

- **Routine Kleihauer–Betke has little value acutely**, but fetal cells $\geq 0.1\%$ predicted contractions/preterm labor.
- **D-negative** → **consider anti-D immunoglobulin** (may omit if FMH test negative). Alloimmunization may still occur if FMH **>15 mL fetal cells** (dose by calculation — Ch 18).
- Check **tetanus** status; give **Tdap** when indicated (neonatal pertussis benefit).

6. Thermal Injury

- Treat as for nonpregnant women; **pregnancy does not alter maternal outcome**. Survival parallels **% body surface area burned**; worse with suicide attempts and **inhalation injury**.

Burn (% TBSA)	Maternal mortality	Fetal mortality
20%	~4%	~20%
40%	~30%	~48%
60%	~93%	~96%

- Severe burns → spontaneous labor within **days to a week**, often stillbirth (hypovolemia, lung injury, sepsis, catabolic state).
- Later pregnancies: painful abdominal **contractures** (release if scars span **>75%** of abdomen); nipple loss affects breastfeeding. Postpartum abdominal skin is a good graft source.

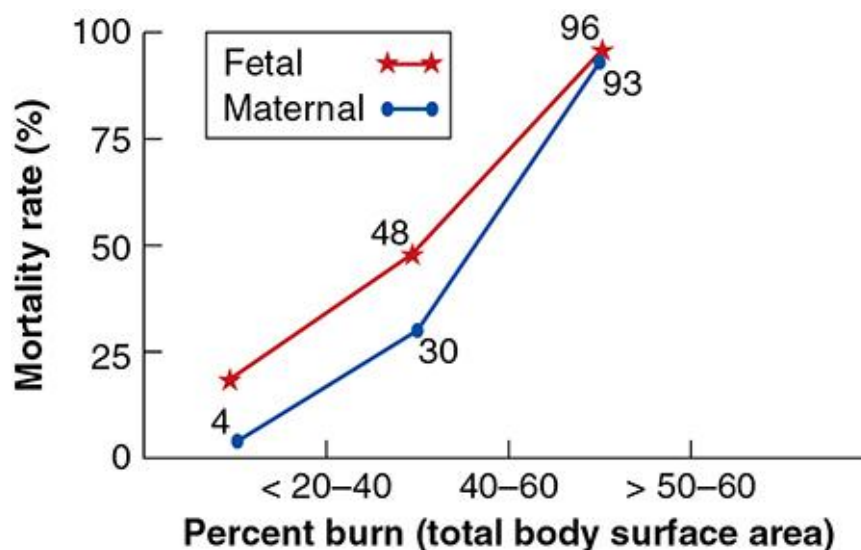


Figure 50-12. Maternal and fetal mortality climb steeply with burn size (211 women): ~4/20%, 30/48% and 93/96% at 20, 40 and 60% TBSA.

Electrical and lightning injuries

- 31 women with electric shock: perinatal outcomes **similar to controls** (110-V North American current likely less dangerous than 220-V European).
- **Lightning: ~50% stillbirth** (13 case reports).

7. Cardiopulmonary Resuscitation

- Maternal cardiac arrest: **1 in 12,000 to 36,000** delivery admissions.

- Most common causes: **hemorrhage, anesthesia, heart failure, amniotic-fluid embolism, sepsis.**

Table 50-8. Possible Etiologies for Cardiac Arrest During Pregnancy

Letter	Category	Causes
A	Anesthesia	High neuraxial block, intravenous local anesthetic, airway complications
A	Accidents	Trauma
B	Bleeding	Atony, abnormal placentation, DIC
C	Cardiovascular	Valvar or congenital heart disease, arrhythmias, ischemia, aortic dissection
D	Drugs	Tocolysis, overdose, anaphylaxis, uterotonics, magnesium sulfate
E	Embolism	Venous, amniotic fluid
F	Fever	Sepsis, viral syndromes, ARDS
G	General	Metabolic abnormalities, hypocalcemia or hyperkalemia with massive hemorrhage
H	Hypertension	Shock, preeclampsia syndrome, HELLP syndrome

ARDS = acute respiratory distress syndrome; DIC = disseminated intravascular coagulation; HELLP = hemolysis, elevated liver transaminases, low platelets. Adapted from Jeejeebhoy, 2015; Soar, 2021; Zelop, 2018.

Memory hook: "A-B-C-D-E-F-G-H" — Anesthesia/Accidents, Bleeding, Cardiovascular, Drugs, Embolism, Fever, General, Hypertension.

AHA standards for CPR in the second half of pregnancy

1. **Left lateral uterine displacement** (relieve vena caval compression)
2. **100% oxygen**
3. **IV access above the diaphragm**
4. Treat hypotension: **SBP <100 mm Hg or <80% of baseline**
5. Review and treat the causes of critical illness early
 - Hand position for compressions is **the same** as in nonpregnant women.
 - External compressions give **~30%** of normal cardiac output — less in late pregnancy (aortocaval compression) → **uterine displacement** by table tilt, **wedge under the right hip**, or **manual displacement to the left**.

Table 50-9. High-Quality Cardiopulmonary Resuscitation in Pregnancy

Components
Place provider hands in lower half of sternum
Displace uterus laterally during resuscitation
Provide rapid chest compressions (100 per minute)
Achieve a compression depth of at least 2 inches
Assure adequate chest recoil between compressions
Minimize interruptions of chest compressions
Avoid prolonged pulse checks (no more than 5–10 seconds)
Resume chest compressions immediately after defibrillating
Switch provider of compressions every 2 minutes to avoid fatigue

Perimortem cesarean delivery

- Maternal survival after arrest **60–70%** (many deaths are out of hospital).

- **AHA / SOAP: deliver within 4–5 min of starting CPR** if the fetus is viable and circulation is not restored. **ACOG: consider from 4 min.**
- Neurologically intact neonatal survival vs arrest-to-delivery interval (Clark 1997):

Arrest-to-delivery	Neurologically intact
≤5 min	98%
6–15 min	83%
16–25 min	33%
26–35 min	25%

- Evidence is weak (Katz 2005: 38 cases, large selection bias). In practice the 4-min goal is rarely met; a "crash" cesarean without anesthesia/equipment could cause a preventable maternal death. Distinguish **perimortem** from **postmortem** cesarean; ethical conflicts are unavoidable.
- **CPR complications:** severe **neurological injury** (most feared), **liver lacerations** (3 of 7 survivors), **Takotsubo cardiomyopathy**.
- **Maternal brain death:** somatic support may continue to reach fetal viability (Ch 63).

8. Envenomation

- Snakes, spiders, scorpions, jellyfish, **hymenoptera** (bees, wasps, hornets, ants). Adverse outcomes stem from **maternal** effects.
- **Venom-specific care:** symptomatic treatment, **antivenom when appropriate**, treat anaphylaxis, fetal assessment.

Quick-Recall: Critical Illness in Pregnancy — Targets and Actions

Problem	Key target / threshold	Action
Pulmonary edema	SpO ₂ > 92% ; BNP <100 rules out, >500 rules in	O ₂ (nonrebreather 10–15 L/min), furosemide 20–40 mg IV , echo; not an indication for cesarean
ARDS	P:F < 200 ; PaO ₂ ≥ 60 (SaO ₂ ≥90%), FiO ₂ <50%, PEEP <15	Treat cause, correct anemia, TV ≤6 mL/kg , intubate when PaCO ₂ 35–40
Fluids	COP–wedge gradient ≤ 4 mm Hg → edema risk	Conservative fluids; albumin no better
Sepsis	Urine ≥ 30 (50) mL/h	2 L (up to 4–6 L) crystalloid , antibiotics ≤ 1 h , lactate, cultures, source control
Septic shock	Hypotension despite fluids	Norepinephrine (or vasopressin); consider low-dose steroids
Group A strep / clostridial gangrene	Uterine hypertonus, capillary leak, shock	Prompt hysterectomy
Blunt trauma	Contractions ≥1 per 10 min → ~20% abruption	Mother first; FHR/contraction monitoring ≥4 h ; anti-D if D-negative
Penetrating abdomen	Fetus injured in %	Laparotomy for gunshot ; selective for stab
Burns	20/40/60% TBSA	Maternal death ~4/30/93%; fetal ~20/48/96%
Cardiac arrest	Delivery ≤5 min → 98% intact	Left uterine displacement, 100 compressions/min, IV above diaphragm, perimortem cesarean at 4–5 min

Key Numbers to Memorize

- Critical care **1–3%** of pregnancies; ICU death risk **2–11%**; ICU transfer **~0.5%**

- $MAP = (SBP + 2DBP)/3$; $SVR = [(MAP - CVP)/CO] \times 80$
- Pulmonary edema **1/500**; 83% hypertensive; β -mimetics once **40%**; furosemide **20–40 mg IV**
- ARDS **60/100,000**, mortality **9%**; P:F **<200**; shunt **>30%** = refractory hypoxemia; proliferative phase **day 7–21**
- 1 g Hb carries **1.25 mL O₂**; P50 **27 (maternal) vs 19 (fetal) mm Hg**; 2,3-DPG **+30%**
- COP **28 → 23 → 17 mm Hg**; 1 g/dL albumin $\approx$ **6 mm Hg**; COP–wedge gradient normally **>8**, danger $\leq$ **4**
- Ventilate: TV $\leq$ **6 mL/kg**, PaO₂ **>60**, PaCO₂ **35–45**; PEEP safe **5–15**; ECMO mortality **~30%**
- Sepsis **2.4–10/10,000**; **12%** of pregnancy-related deaths; septic shock mortality **20%**; each failed organ **+15–20%** death risk; GAS/clostridial toxin mortality **58%**
- Trauma **10–20%** of pregnancies; MVA + falls **85%**; IPV **1 in 5** women; sexual assault PTSD/depression **30–35%**
- Abruption **1–6%** minor, $\leq$ **50%** major, **>30 mph**; FMH **<15 mL** in **90%**; uterine rupture **<1%** (25 mph → 500 mm Hg)
- Penetrating: maternal visceral injury **15–40%**; maternal death **7%**, fetal **73%**
- Burns 20/40/60% → maternal **4/30/93%**, fetal **20/48/96%**; lightning stillbirth **50%**
- Arrest **1/12,000–36,000**; survival **60–70%**; compressions give **30%** CO; perimortem cesarean at **4–5 min**; $\leq$ 5 min → **98%** intact